

- 8 (a) Simple harmonic motion is defined as the motion of a particle (or body) about a fixed point such that its acceleration is proportional to its displacement from the fixed point and is always directed towards the point.
- (b) (i) $M = AL\rho$
- (ii) $\Delta p = h\rho g = 2xA\rho g$
 $\Delta p = \frac{F}{A} \Rightarrow F = (\Delta p)A = 2xA\rho g$
- (iii) Since F is the restoring force, (F acts in the opposite direction to x)
 $F = -2xA\rho g$
 $F = Ma$ where a is the acceleration
 $\therefore -2xA\rho g = AL\rho \cdot a$
 $\therefore a = \frac{-2xA\rho g}{AL\rho} = \frac{-2gx}{L}$
 $\therefore a \propto -x$, since $\frac{2g}{L}$ is a constant
Hence the motion is s.h.m.
- (iv) Since $a = -\omega^2 x$ and $a = -\frac{2gx}{L}$
 $\therefore \omega^2 = \frac{2g}{L} \Rightarrow \omega = \sqrt{\frac{2g}{L}}$
- (c) (i) $\omega = \sqrt{\frac{2g}{L}} = \sqrt{\frac{2 \times 9.81}{0.92}} = 4.62 \text{ rad s}^{-1}$
- (ii) $T = \frac{2\pi}{\omega} = \frac{2\pi}{4.618} = 1.36 \text{ s}$
- (iii) $E_{K,\max} = \frac{1}{2} M v_{\max}^2 = \frac{1}{2} M \omega^2 x_0^2$
 $= \frac{1}{2} (1.2) (4.618)^2 (0.080)^2$
 $= 0.0819 \text{ J}$
- (d) (i) Damping refers to the loss of energy in an oscillatory system (due to the presence of dissipative forces).

(ii)

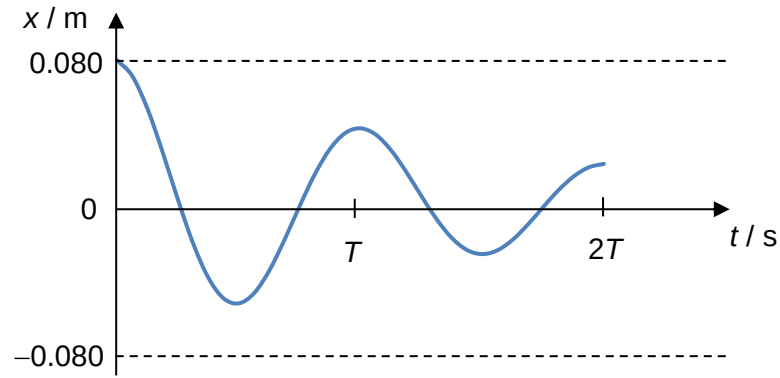


Fig. 8.3

(iii)

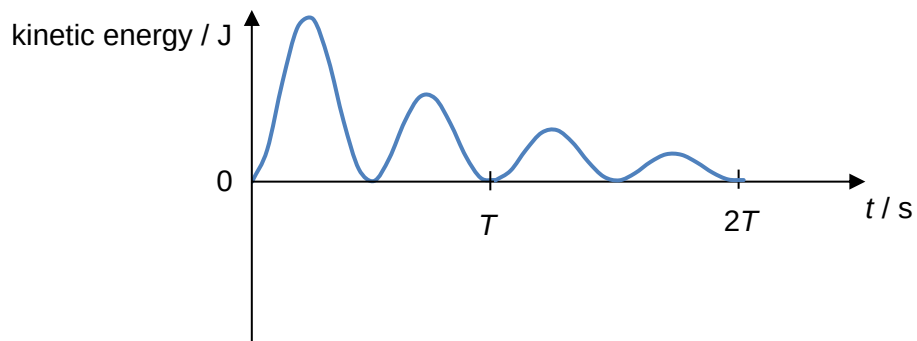


Fig. 8.4

- (e) Car suspension,
to reduce motion sickness of passengers
OR damping of oscillation of pointer in analogue meters, to allow readings to be taken more quickly
OR door closer, to close door without door slamming or swinging.
OR mass dampers in buildings